PRINCETON SCIENCE OLYMPIAD

2021 Division C
Water Quality Test

Instructions:

- This exam has 63 questions for a total of 116 points.
- Teams of up to 2 will have 50 minutes to complete the exam.
- Each team may use one 8.5" x 11" sheet of paper with information from any source and two non-programmable, non-graphing calculators. Other sources, including the internet, are not permitted during the exam.
- *Due to the online format of this exam, a participant-built salinometer is no longer required for this event.*

Team number: _____
Team Name: _____
Names: _____
_____

Point Total: ____/116

Part 1. Estuary and Marine Ecology (31 points)

1. What property of water does the water strider take advantage of to walk on water, as in the image below? What is the primary molecular force responsible for this property? (2 points)



Figure 1. *An insect commonly known as a water strider is shown here on the surface of water.*

2. While most ecological pyramids are largest at the level of the _____, the _____ pyramids of some marine ecosystems are inverted. (1 point)
    a. producer; biomass
    b. apex predator; biomass
    c. apex predator; numbers
    d. producer; numbers
3. Recalling your answer to question 2, explain why some pyramid representations of marine ecosystems are inverted. (1 point)
4. The shape and movement of garbage patches like the Great Pacific Garbage Patch are largely controlled by systems of currents known as _____. (1 point)
5. Marine animals that are unintentionally captured by fishing operations—such as dolphins, whales, and sea turtles—are known as _____. (1 point)
6. Common estuary classification schemes are based on (choose all that apply, 2 points):
    a. geology
    b. water circulation
    c. flora & fauna
    d. size
7. The reproducible arrangement of estuarine species into "zones" primarily reflects differential adaptations to (1 point):
    a. Environmental gradients (e.g. salinity, light)
    b. Food web structure
    c. Successional stages (e.g. primary, secondary)
    d. Nutrient pollution

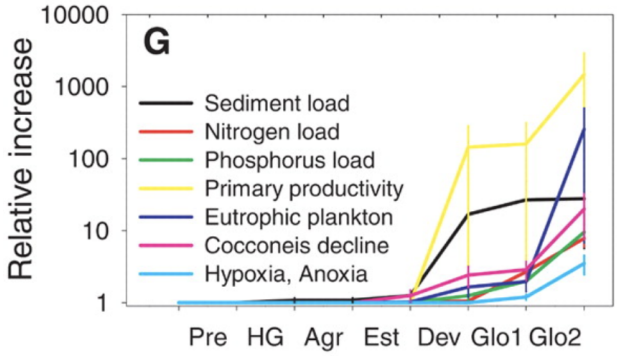


Figure 2. Refer to this figure for questions 8-10. Note that the x-axis represents time throughout human cultural periods from prehistory (Pre) to the present (Glo2). The y-axis represents the relative increase in a set of relevant water quality indicators, labeled by color.

8. The changes in global estuarine water quality (Figure 2) indicate widespread _____, a process characterized by trends including hypoxia, anoxia, and algal blooms (1 point).
9. Diatoms of the genus *Cocconeis*, classified as **epiphytes**, are often found in seagrass meadows. How does the process identified in question 8 drive their decline? Is this phenomenon propagated through food webs by bottom-up or top-down processes? (3 points, **TIEBREAKER**)
10. Provide an example of a point and non-point pollution source likely to directly contribute to the phenomenon identified in question 8. Which type of pollution is more difficult to control? (3 points)
11. _____ adaptations to environmental challenges are heritable. (1 point)
    a. Physiological
    b. Evolutionary
12. Provide three examples of adaptations of estuarine organisms, including the environmental challenge(s) in estuaries each adaptation responds to (3 points).
13. Some mangrove trees stock their tissues with solutes like proline and mannitol. What characteristic of estuaries does this adaptation respond to? (1 point)
    a. anoxia
    b. hyperoxia
    c. high acidity
    d. high salinity
14. Explain why the condition identified in question 13 would be a problem for most terrestrial tree species without similar adaptations. What process of membrane-separated aqueous solutions mediates this phenomenon? (2 points)

15. *Vibrio* bacteria are endemic in some marine habitats and estuaries, and can cause enteric and other life-threatening illnesses in humans. Based on this information, are outbreaks of *Vibrio spp.* likely to coincide with outbreaks of other pathogens like *Salmonella* and *Campylobacter* bacteria from estuary and marine environments? Why or why not? (3 points)
16. What type of estuary organism is a common reservoir of human pathogens? Why? (2 points)
17. How do seabirds mediate the phosphorus cycle? (3 points)

Part 2. Coral Reef Macroflora and Fauna (45 points)

18. (True/False) The primary productivity of coral reefs is around twice as high as the primary productivity of the open ocean (1 point).
19. Name the largest coral reef system in the world (1 point).
20. Most corals are classified as (r-selected/K-selected) (1 point).
21. Name a type of reef-building organism <u>other than</u> stony coral. (1 point).

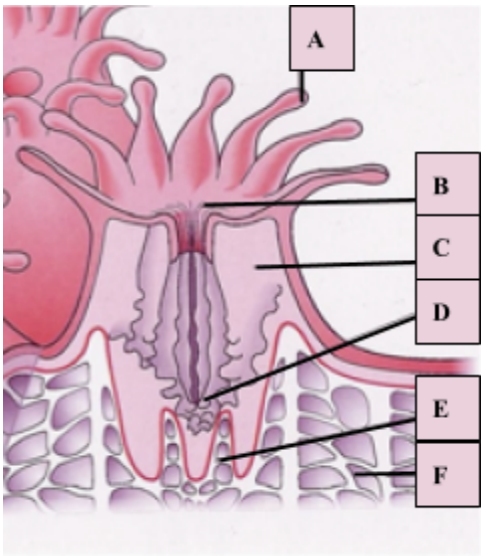


Figure 3. Refer to the diagram of a coral polyp cross section above to answer questions 5-8. For 5-7, choose the letter corresponding to the appropriate structure in the image.

22. Which structure contains nematocysts? (1 point)
23. Which structure expels waste products? (1 point)
24. Which structure contains gonads? (1 point)
25. Name two purposes of structure A (2 points).

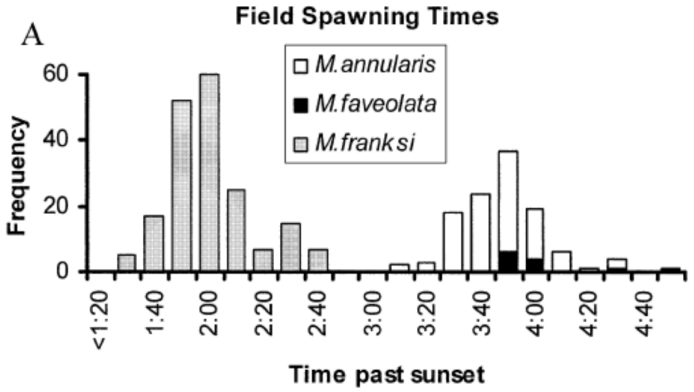


Figure 4. For questions 9-13, refer to this figure, which tracks spawning times of three species of coral in the genus *Montastraea*.

26. _____ spawning occurs when corals release gametes into the water column for external fertilization. (1 point)
27. The type of spawning identified in question 9 is a form of _____ reproduction (1 point).
    a. sexual
    b. asexual
28. Based on figure 4, a gamete of the species *M. franski* would have the greatest chance of successful fertilization at which time past sunset (1 point):
    a. 1:40
    b. 2:00
    c. 2:20
    d. 2:40
29. Based on Figure 4, predict which pair of species would be least likely to have compatible gametes (i.e., gametes that can fertilize each other). Explain your prediction (3 points, **TIEBREAKER**).
30. Explain how ocean warming could threaten coral species that rely on this type of spawning to reproduce (2 points).
31. Most stony coral pigmentation is provided by dinoflagellates known as _____. When these organisms are lost or expelled under environmentally stressful conditions, corals lose their color in a process known as _____ _____. (2 points)
32. Choose all of the below required for coral reef formation (1 point):
    a. clear water
    b. high nitrogen content
    c. solid structural base for attachment
    d. temperatures from 20-27˚C
    e. low sulfur content

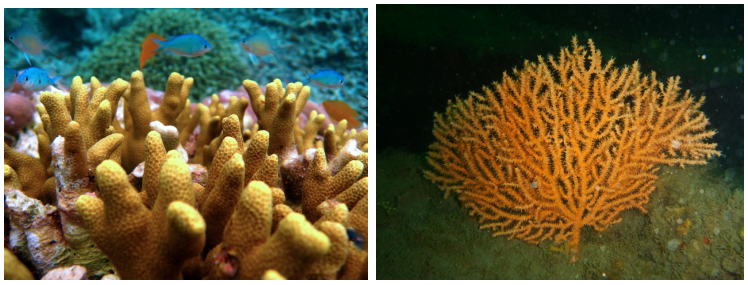


Figure 5. Refer to this figure for question 16.

33. Explain the basic difference between calcium carbonate storage in the corals shown at left and at right in Figure 5 (2 points).
34. The chemical formula of calcium carbonate is _____. The common crystal form of calcium carbonate deposited by hard corals is known as _____ (2 points).
35. The structure of a typical coral reef food web is outlined in Figure 6 below. For question 14 parts a-e, provide an example of a coral reef organism in each corresponding ecological category labeled A-E in the figure. Common names are acceptable and species-level identification is not necessary. (5 points)

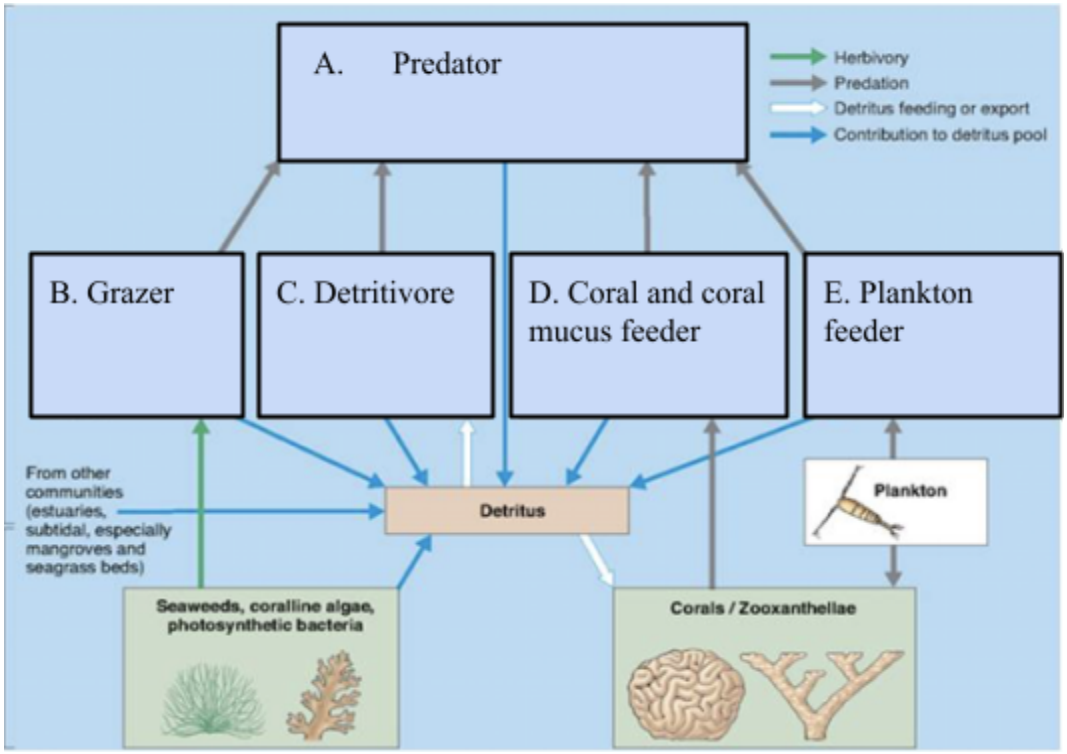


Figure 6. Refer to this figure for parts a-e.

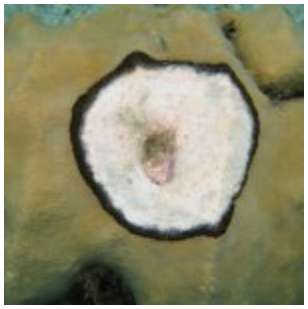
a. Predator:
b. Grazer:
c. Detritivore:
d. Coral and coral mucus feeder:
e. Plankton feeder:

36. Many coral reef restoration efforts grow new corals through methods involving a type of asexual reproduction. Briefly describe these methods. (2 points)
37. Explain one major advantage and one major disadvantage of the restoration method in question 15 compared to restoration projects involving sexual reproduction of corals (2 points).

For questions 17-19, state whether or not the coral structure shown appears healthy or unhealthy. If unhealthy, identify and describe (in no more than 15 words) any apparent mortality or disease. (3 points)

38. 

39. 

40. 

41. Explain the coral-algal phase shift phenomenon and its implications for the biodiversity of coral reef ecosystems (4 points).



Figure 7. Refer to this figure for question 25.

42. Give the common name and genus of the fish pictured in Figure 6. How would you expect populations of this fish to react to the coral-algal phase shift phenomenon? (2 points)

Part 3. Water monitoring and analysis (29 points)

43. Provide an aragonite saturation state value that would be considered corrosive to shells, including units (1 point).
44. How does ocean acidification affect the mean aragonite saturation state value of marine environments? (1 point)
    a. No effect
    b. Increases it
    c. Decreases it
45. Explain how a secchi disk is used, including what it measures (2 points).
46. Compare and contrast how dissolved oxygen (DO) and biological oxygen demand (BOD) are measured (2 points).
47. Explain the relationship between turbidity and total suspended solids (TSS); what does each measure and which, if either, is more useful for determining water quality? (2 points)

Table 1. The following average measurements were taken in the freshwater of the Scioly River in the midwestern United States during the years indicated. Questions 6-9 are based on these measurements.

Year	**Specific conductance** (µS/cm)	**pH**
1953	62	6.4
1983	95	6.5
2013	131	6.8

48. What property of water is specific conductance often used to assess? (2 points)
    a. Salinity
    b. Fecal coliform count
    c. Dissolved oxygen
    d. Temperature
49. Are the trends in specific conductance and pH (Table 1) likely to be related? Why or why not? (3 points, **TIEBREAKER**)
50. After observing these trends in the Scioly River (Table 1), diligent Science Olympiad students find similar trends in five more rivers across the country. Provide a possible explanation for these trends (2 points).
51. Should the students be concerned by these trends? Why or why not? (2 points).
52. A dam releases cold water from the lower levels of a water reservoir. Is this considered thermal pollution? (1 point)
    a. Yes, and it is likely to impact downstream ecosystems.
    b. Yes, but it is not likely to impact downstream ecosystems.
    c. No, and it is not likely to impact downstream ecosystems.
    d. No, but it may impact downstream ecosystems.

Questions 53-56 below refer to fecal coliform measurements (6 points).

53. How is fecal coliform measured?
54. True/False: Fecal coliform bacteria typically cause disease in humans.
55. You find a fecal coliform count of 97 colonies/100 mL. Should you recommend taking additional steps to disinfect the water? Justify your answer.
56. Which of the following CANNOT be used to lower fecal coliform counts?
    i. filtration
    ii. phosphates
    iii. UV treatment
    iv. iodine
57. Excess _____ are often toxic to organisms including humans, in which these compounds can interfere with oxygen transport (1 point).
    a. phosphates
    b. nitrates
58. Why do fish die-offs often happen in the summer? (2 points).
59. Gases including CO_2, H_2S, and HCl constantly bubble up from magma chambers underneath Yellowstone's geysers. Why do some of the geysers have pH values above 7.0? (2 points)

Part 4. Salinometer (11 points)

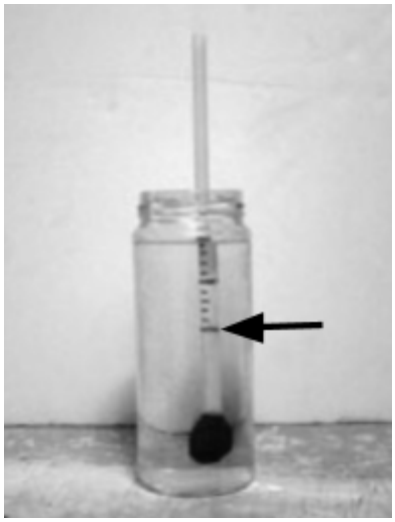


Figure 8. Refer to this figure for questions 1-3 below. The salinometer is indicated by the arrow.

60. The type of salinometer shown in figure 8 is also known as a _____ (1 point).
61. Explain how the salinometer in figure 8 can be used to measure salinity, including the key property of water it relies on and how it would be affected by the addition of more salt to the solution above (5 points).
62. Why do salinometers like the one in figure 8 need to be calibrated before use? (2 points)
63. Describe another major type of salinometer that measures salinity using a different key property of water. In your answer, make sure to explain how the key property you identified changes with salinity and identify the key part(s) of this type of salinometer. (3 points)